

# SHIVANI RAKAM

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## SUMMARY

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Curious and impact-driven, I work at the intersection of data science, engineering, and analytics to solve problems that matter. From building real-time data pipelines to developing predictive models and dashboards, I enjoy simplifying complexity and enabling smarter decisions. I've delivered solutions across finance, healthcare, and product domains—always focusing on how data can accelerate business outcomes and empower people.

## EXPERIENCE

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### Data Science Engineer – JPMorgan Chase | USA

*JUL 2024 – Present*

- Tasked with enhancing fraud detection workflows, I engineered PySpark pipelines on Databricks to automate real-time data ingestion and transformation, cutting processing time significantly.
- Designed and deployed machine learning models using XGBoost and Random Forest, improving fraud detection precision and reducing false positives by 25%.
- Led the setup and monitoring of ML experiments using MLflow, ensuring traceable and auditable model development aligned with regulatory standards.
- Collaborated with DevOps teams to implement CI/CD pipelines via Jenkins, reducing model deployment errors and accelerating release cycles.
- Delivered actionable insights through ad hoc analyses for compliance teams, directly shaping strategic risk decisions.
- Presented insights and roadmaps to executives, supporting data-informed enhancements that improved campaign performance by 15%.
- Collaborated with engineering teams to validate API-based data flows and troubleshoot data issues using Postman.
- Worked within an Agile environment to deliver iterative data science features through biweekly sprints and sprint reviews.

### Data Analyst – State Street | USA

*JUL 2023 – MAY 2024*

- Developed and maintained scalable ETL pipelines using SQL and Python to automate financial reporting workflows and improve data reliability.
- Designed and optimized Power BI dashboards and Looker reports for executive leadership, enabling real-time monitoring of KPIs and financial health.
- Conducted ad hoc and exploratory analyses to identify operational inefficiencies, resulting in a 12% improvement in resource utilization.
- Supported product analytics by creating customer segmentation models and A/B test evaluations for marketing and strategy teams.
- Collaborated cross-functionally with engineering, finance, and compliance to align data architecture and business logic across systems.
- Utilized Snowflake to manage and query large-scale financial data efficiently for reporting and analytics workflows.
- Participated in Scrum ceremonies including daily stand-ups, sprint planning, and retrospectives to ensure alignment on analytics initiatives.

### Data Analyst – Infosys | Hyderabad, India

*JAN 2021 - JUL 2022*

- Built end-to-end data pipelines using PySpark and SQL to process healthcare claims and EMR data for downstream analytics and reporting.
- Developed logistic regression and classification models to predict readmission risk and support care planning decisions.
- Automated data quality checks and job scheduling using Shell scripting and Autosys, improving uptime and reducing manual effort.
- Created departmental dashboards using Power BI to visualize patient trends, costs, and compliance metrics.

- Partnered with data engineers and clinical stakeholders to validate model features and improve reporting logic across multiple regions.
- Identified and measured KPIs related to patient readmission rates, provider performance, and claims processing efficiency, helping clinical teams focus on measurable outcomes.

## EDUCATION

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### George Mason University

*Aug 2022 - May 2024*

*Master's, Information Systems GPA: 3.8/4*

**Relevant coursework:** Database systems, Information Systems Analysis and Design, Analytics Big Data to Information, Rapid Information Systems Prototyping, Software Project Management, Info: Represent, Process, Viz, Discrete/Logic Structure for INFS, Cloud Computing Security, Principles and Practices of Communication Networks, Cyber Security Fundamentals.

### SR Engineering College

*Aug 2018 – May 2022*

*Bachelor's, Electronics in Communication Engineering GPA: 8/10*

## SKILLS

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- **Programming Languages:** Python, R, C++, Java, Scala, SQL, Shell Script, MATLAB
- **Big Data Tools:** Apache Spark, Apache Airflow, Kafka, Databricks, Hive, Hadoop, Cassandra
- **ETL/ELT Tools:** DBT, Matillion, Informatica, SSIS
- **Databases:** SQL Server, MySQL, Snowflake, PostgreSQL, NoSQL, Oracle, MongoDB, DB2
- **Business Intelligence & Visualization:** Power BI, Tableau, Looker Studio
- **DevOps & Collaboration Tools:** Terraform, Git, GitHub, Bitbucket, Jenkins, Docker, Jira, Confluence, VS Code
- **Cloud Computing:** AWS (S3, CloudWatch, Athena, RedShift, EMR, EC2, IAM, Secret Manager, Lambda, SNS, SQS, Glue; Azure (Azure Data Factory, Azure Databricks, Blob Storage), GCP – Big Query
- **Machine Learning:** Classification, Regression, KNN, Decision Trees, Random Forests, XGBoost, Time Series (ARIMA, FB Prophet), Ensemble models, A/B Testing
- **ML Tools/Framework:** Scikit-learn, PyTorch, Keras, TensorFlow, Spark
- **Methodologies:** Agile, Scrum, CI/CD, Data Modeling, Data Governance, Unit Testing

## CERTIFICATION & TRAINING

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- AWS Cloud Practitioner (AWS, 2024)
- AWS Solutions Architect (Udemy, 2024)
- Tableau Desktop Specialist (Salesforce, 2024)
- SQL for Data Analysis (Udacity, 2025)
- Advanced SQL (HackerRank, 2025)